

$\downarrow$ LHS $\rightarrow$ creation of array
 $\rightarrow$ `int arr = new int[5];`
 $\rightarrow$ `new int[6][i, j];`
 $1+3 \neq \text{target}$
 $1+4 \neq$
 $\text{nums} = [1, 3, 4, 2, 6]$ $\text{target} = 3$
 $\rightarrow$ `int arr[] = new int[2];`
`for (int i = 0; i < n; i++)`
`for (int j = i + 1; j < n; j++)`
`if (nums[i] + nums[j] == target)` $[0, 0]$
 $\rightarrow \text{arr}[0] = i$
 $\rightarrow \text{arr}[1] = j;$ $[0, 3]$
`return arr;`

`for (int i = 0; i < n; i++)`
`for (int j = i + 1; j < n; j++)`
 $\uparrow$
 $\text{depends on } i$
 $\rightarrow$ for each of value of i
 $\rightarrow$ It is running.
 $\rightarrow$ worst case $\rightarrow n$ times.
 $n \times n = n^2$
 $n + n + n + \dots + n$
 $n \text{ times}$
 $= n^2$
 $i = 0 \rightarrow n$
 $i = 1 \rightarrow n-1$
 $i = 2 \rightarrow n-2$
 $\vdots$
 $i = n-1 \rightarrow 0$
 $0 + 1 + 2 + \dots + n-1 + n$

`for (int i = 0; i < n; i++)`
`for (int j = 0; j < n; j++)`
`Sout(" ");`
`}`
`}`
 $i = 0 \rightarrow j = 0 \rightarrow n$ (n times)
 $i = 1 \rightarrow j = 0 \rightarrow n$ (n times)
 $i = 2 \rightarrow j = 0 \rightarrow n$
 $\vdots$
 $i = n \rightarrow j = 0 \rightarrow n$
 $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow \dots \rightarrow n$
 $n \text{ times}$
 $i = 1 \rightarrow n$
 $i = 2 \rightarrow n$
 $i = 3 \rightarrow n$
 $\vdots$
 $i = n \rightarrow n$
 n^2

$4 + 4 + 4 + 4 = 16 \rightarrow 4^2$

$n + n \rightarrow n \times 2$

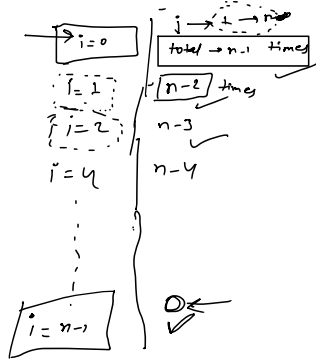
$$n+n+n \rightarrow n \times 3$$

$$n+n+n+n \dots n \rightarrow n \times n$$

for (int i=0 → n)

↳ for (j=i+1 → n)

$$0+1 \rightarrow n-1$$



$$0 \quad i=1 \quad i=2 \quad i=n-1$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$n-1 + n-2 + n-3 + \dots + 0$$

$n=2$ Natural number

$$1, 2$$

$$1+2 = 3$$

$$\frac{n \times (n+1)}{2}$$

$$O\left(\frac{n \times (n+1)}{2}\right)$$

$$\frac{2 \times (3)}{2} = 3$$

$$O\left(\frac{n^2+n}{2}\right) \rightarrow O(n^2) = \boxed{O(n^2)}$$

10^8 → threshold

↳ algorithm should not exceed.

$$n=10^4$$

$$O(n^2)$$

$$n^2 = (10^8)$$

$$n = 3 \times 10^4$$

$$O(n^2) \times$$

$$n^2 = 9 \times 10^8$$

$$10^8$$

$$n = 9 \times 10^4$$

$$n \log n$$

T.C	↑ Inc. or dec
$O(n^4)$	
$O(n^3)$	
$O(n^2 \log n)$	
$\rightarrow O(n^2) \times$	
$\rightarrow O(n \log n)$	
$O(n)$	
$O(\log n)$	

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$$= 9 \times 10^4 \times \log_2(9 \times 10^4)$$

$$9 \times 10^4 \times (\log_2 9 + \log_2 10^9)$$

$$9 \times 10^4 \times (3. \cancel{\dots} + 4 \times 3. \cancel{\dots})$$

$$9 \times 10^4 \times 15$$

$$\hookrightarrow 135 \times 10^4$$

$1.35 \times 10^6 \longrightarrow 10^8$

target = 2

eg:-

target = 7

array: [1, 2, 3, 4, 7, 11, 15, 18]

1 → smallest
4 → target
18 → largest

So, test ✓

target = 1/4

$$O(n^2) \times$$

$$1 + 1^2 \neq 14$$

$11 + 2 = 13 < \text{target}$

$19 > \text{target}$

$a + b$ $> target$
↓

$12 + 3 = 14 \Rightarrow \text{forget}$

$$\text{arr}[0] = i+1;$$

$$a_{rr} [1] = j+1;$$

$$1 + 11 = 12$$

↑ 12 < target

while (i, j)